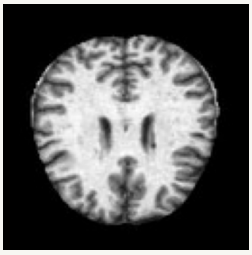
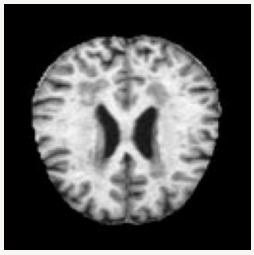
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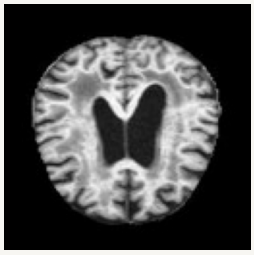
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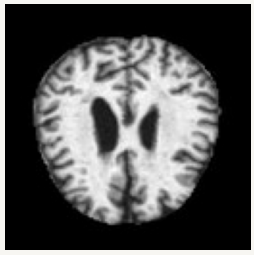
Alzheimer’s Disease Stage Classification from Brain MRI with Deep Learning

ANN vs. CNN vs. Transfer Learning — and why 97% accuracy can be misleading: a data-leakage analysis


Non Demented


Very Mild


Mild


Moderate

Prepared by	**Zeynep Erkişi**
Student ID	**220302027**
Course	**Deep Learning**
Submission date	**27 November 2026**

Alzheimer's Disease Stage Classification from Brain MRI with Deep Learning

Zeynep Erkişi · Student ID 220302027 · Istanbul Arel University · Deep Learning – Final Project Report · Submission date: 27 November 2026

Abstract. We classify 2D brain MRI slices into four Alzheimer's disease stages using 6,400 images from an ADNI-derived public dataset. We compare an ANN, a CNN trained from scratch and an ImageNet-pre-trained EfficientNetB0, all with class-weighted loss. On the official test split the ANN reaches 96.9% accuracy, far ahead of both convolutional models. A nearest-neighbour test explains this result: 98.9% of test images have a near-identical training image ($r > 0.95$), and a 1-NN lookup scores 99.8%. When data augmentation blocks this memorisation, macro-F1 falls to 0.34–0.54 and the models rank as expected: EfficientNetB0 > CNN > ANN > majority baseline. The study shows that high benchmark scores in medical imaging must be checked for data leakage.

1. Problem

The task is to classify a single 2D brain MRI slice into one of four stages of Alzheimer's disease: **Non Demented**, **Very Mild Demented**, **Mild Demented** and **Moderate Demented**. It is a supervised, four-class image classification problem.

Why it matters

More than 57 million people live with dementia, and Alzheimer's disease causes an estimated 60–70% of cases (source). The disease damages brain tissue years before symptoms are obvious. On MRI this appears as shrinking of the cortex and hippocampus and enlarged ventricles (the dark fluid-filled cavities in the centre of the brain).

Reading these changes takes an expert radiologist, and the early stages are subtle. A model that flags likely early-stage cases could help prioritise patients for specialist review. Earlier diagnosis gives patients more time to plan and to join treatment trials.

Why it is interesting for deep learning

- **Subtle visual differences.** Neighbouring stages (e.g. Non Demented vs. Very Mild) differ by small changes in tissue shape, not by obvious objects. This is a good test of what convolutional features can learn.
- **Severe class imbalance.** The most important clinical classes are also the rarest. Only 1% of images are Moderate Demented, so plain accuracy is misleading.

- **Comparing model families.** The course covers ANNs, CNNs and pre-trained networks. This task lets us compare all three on the same data.
- **Trustworthiness.** Medical models must be checked for shortcuts. Many published Alzheimer's MRI studies report very high accuracies that later turned out to come from data leakage, where slices of the same patient appear in both training and test sets (Wen et al., 2020; Yagis et al., 2021). We test for this effect directly in our own experiments.

Research questions

1. How well do an ANN, a CNN trained from scratch and a pre-trained EfficientNetB0 classify Alzheimer's stages?
2. How much does each model beat a trivial baseline that always predicts the most common class?
3. Are the reported test scores realistic, or are they inflated by near-duplicate images shared between training and test sets?

2. Data

We use the **Alzheimer MRI dataset** published on Hugging Face as Falah/Alzheimer_MRI (Apache-2.0 licence). It re-packages the widely used Kaggle *Alzheimer MRI Preprocessed Dataset*, whose images originate from the ADNI study. It contains **6,400 grayscale axial MRI slices of 128 × 128 pixels** (28 MB) and comes with a fixed train/test split.

Class	Train	Test	Total	Share
Non Demented	2,566	634	3,200	50%
Very Mild Demented	1,781	459	2,240	35%
Mild Demented	724	172	896	14%
Moderate Demented	49	15	64	1%
Total	**5,120**	**1,280**	**6,400**	100%

Each sample is a pair *(image, label)*: a 128 × 128 array of 8-bit intensities and an integer class. The original label ids (0 = Mild, 1 = Moderate, 2 = Non, 3 = Very Mild) were re-ordered by severity for readability. Figure 1 shows five random slices per class. The visible differences are subtle: larger ventricles and more widened grooves on the brain surface in the demented classes.

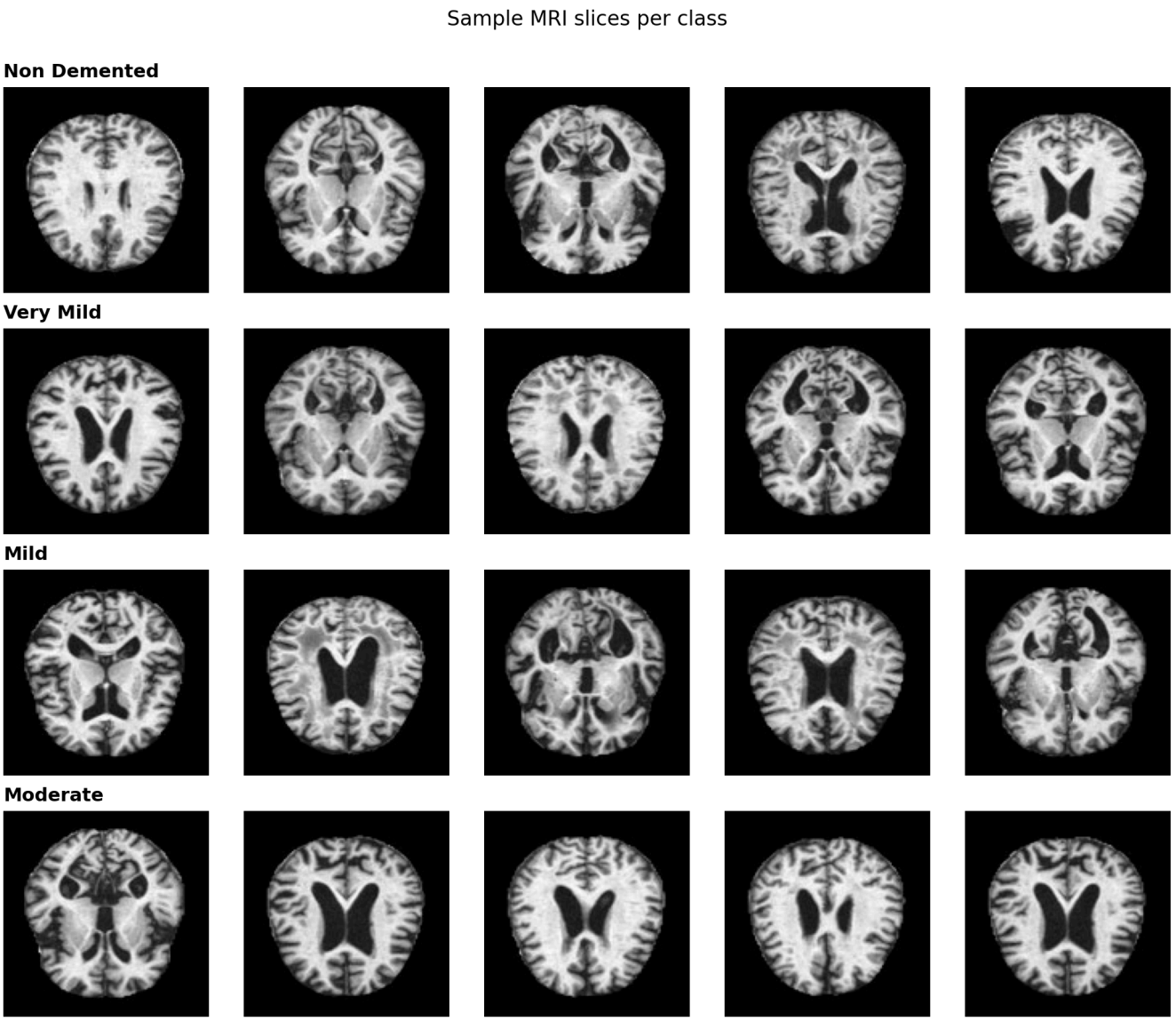


Figure 1. Five random training slices per class.

Which parts of the data are used

- **Training set (4,352 images).** 85% of the official training split, used to fit the weights.
- **Validation set (768 images).** The other 15%, drawn with a stratified split (seed 42). It is used only for early stopping and learning-rate scheduling.
- **Test set (1,280 images).** The official test split, used once, for the final evaluation only.
- **Pre-processing.** Pixel values are scaled to [0, 1]. For EfficientNetB0 the single channel is copied to three channels, because the network was pre-trained on colour images.

Potential obstacles

1. **Class imbalance.** Moderate Demented has only 49 training and 15 test images. A model can reach 50% accuracy by always answering Non Demented. We counter this with **class-weighted loss** (weights ≈ 0.50, 0.72, 1.77 and 26 for the four classes) and report

imbalance-aware metrics.

2. **Very small minority class.** With 15 test images, one extra correct Moderate prediction moves its recall by about 7 points, so per-class results for this class are noisy.
3. **Near-duplicate slices (data leakage).** The dataset holds several neighbouring slices per subject but no patient ids, and the official split is per image. Slices of the same brain can therefore land in both training and test sets. This would make the test set easier than truly new patients. Section 4.5 measures this directly.
4. **Limited input.** Each image is one 2D slice, down-sampled to 128 × 128. The full 3D scan and clinical data (age, cognitive scores) are not available.

3. Methods

We compare three models of increasing power, each trained on the same split with the same loss. A trivial majority-class predictor serves as the floor. All code is in Python with TensorFlow/Keras and runs in Google Colab on a T4 GPU.

Model	Architecture	Parameters	Data augmentation
A. ANN (MLP)	Flatten → Dense 512 → Dense 256 → softmax; BatchNorm + Dropout 0.4 after each hidden layer	8.52 M	No
B. CNN from scratch	4 blocks of (Conv 3×3 → BN → ReLU) × 2 + MaxPool, 32→64→128→256 filters; global average pooling; Dense 128; Dropout 0.4/0.3	1.21 M	Yes
C. EfficientNetB0 (transfer learning)	ImageNet-pre-trained EfficientNetB0 backbone + global average pooling + Dropout 0.4 + softmax	4.05 M	Yes

Model A – ANN baseline

The multilayer perceptron sees the image as a flat vector of 16,384 numbers. It has no notion of neighbouring pixels, so it shows what can be learned without any spatial structure. It is the classic first model from the course.

Model B – CNN from scratch

Convolutions share weights across the image and build features from edges up to shapes. This suits anatomical changes such as larger ventricles. Global average pooling keeps the parameter count low (1.2 M vs. 8.5 M for the ANN), which matters with only 4,352 training images.

Model C – Transfer learning with EfficientNetB0

We reuse the existing Keras implementation of EfficientNetB0 (Tan & Le, 2019) with ImageNet weights. Training has two phases:

1. **Feature extraction.** The backbone is frozen and only the new classification head is trained (Adam, learning rate 1e-3, up to 15 epochs).
2. **Fine-tuning.** The top 40% of backbone layers are unfrozen and trained with a 100× smaller learning rate (1e-5, up to 40 epochs). BatchNorm layers stay frozen to keep their pre-trained statistics.

Several checkpoints fine-tuned on this dataset are public on Hugging Face (e.g. ResNet and ViT models). We do not copy them. Instead we start from generic ImageNet weights and add two things they lack: class-weighted training and a leakage check.

Training setup (all models)

- **Loss.** Sparse categorical cross-entropy, weighted per class with sklearn's *balanced* weights.
- **Optimiser.** Adam, batch size 32.
- **Regularisation.** Dropout, batch normalisation, early stopping on validation loss (patience 10; 5–8 for EfficientNet phases) with best-weight restore, and learning-rate halving on plateaus.
- **Augmentation** (Models B and C). Random horizontal flip, rotation up to ±18°, zoom up to ±10% and shifts up to ±5%, applied on the fly and only during training.
- **Reproducibility.** Fixed random seed 42 for Python, NumPy and TensorFlow.

Leakage test and ablation

The first run produced an unexpected ranking: the ANN scored far above both convolutional models (Section 4). We designed two extra experiments to explain it.

1. **1-nearest-neighbour test.** Each test image receives the label of the training image it correlates with most (Pearson correlation of raw pixels). This classifier learns nothing. If it is nearly as accurate as the best network, the test set contains near-copies of training images.

2. **Augmentation ablation.** Each model is trained both with and without augmentation: ANN + augmentation, CNN without augmentation and EfficientNetB0 without augmentation. Augmentation prevents a model from matching exact pixel patterns, so it separates memorisation from real generalisation.

4. Evaluation and Results

Quantitative metrics. Half of the test set is one class, so accuracy alone is misleading. We therefore report five metrics on the untouched test set:

- **Accuracy.** The share of correct predictions.
- **Balanced accuracy.** Mean recall over the four classes.
- **Macro-F1.** Mean F1 over the classes, our main metric because it treats every stage equally.
- **Cohen's κ.** Agreement with the true labels beyond chance.
- **Macro ROC-AUC (one-vs-rest).** How well the predicted probabilities rank the classes.

We also report per-class precision and recall, and **McNemar's test** to check whether two models differ by more than chance.

Qualitative outputs. We inspect training curves, confusion matrices, Grad-CAM heat-maps (Selvaraju et al., 2017) and misclassified images.

Hypothesis before the experiments. We expected EfficientNetB0 > CNN > ANN > majority baseline. Pre-trained features and convolutions should both help on subtle anatomical changes.

4.1 Main results

Model	Accuracy	Balanced acc.	Macro-F1	Cohen's κ	ROC-AUC	Params	Train time
Majority class	0.495	0.250	0.166	0.000	–	–	–
ANN	**0.969**	**0.972**	**0.961**	**0.949**	**0.997**	8.52 M	44 s
CNN (+ aug.)	0.507	0.422	0.393	0.241	0.782	1.21 M	350 s
EfficientNetB0 (+ aug.)	0.588	0.529	0.541	0.335	0.818	4.05 M	329 s

All three networks beat the majority baseline, but the ranking is the **reverse of our hypothesis**. The simple ANN reaches 96.9% accuracy and 0.961 macro-F1. The two convolutional models stay near 0.4–0.54 macro-F1. Of the two, the pre-trained EfficientNetB0 is clearly better than the CNN trained from scratch (+0.15 macro-F1).

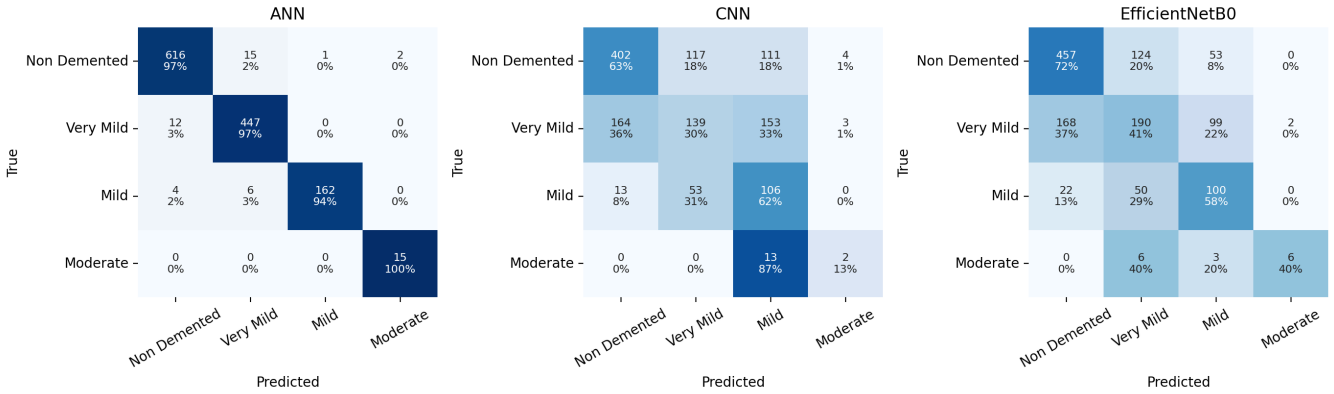


Figure 2. Test-set confusion matrices (counts and row percentages).

The confusion matrices show where the models fail. The ANN makes only 40 errors out of 1,280 and finds all 15 Moderate cases. The CNN and EfficientNetB0 mostly confuse **neighbouring stages**. Very Mild is the hardest class: the CNN recognises only 30% of Very Mild cases and EfficientNetB0 41%, often calling them Non Demented or Mild. This matches clinical reality, since the boundary between normal ageing and very mild dementia is subtle on a single slice.

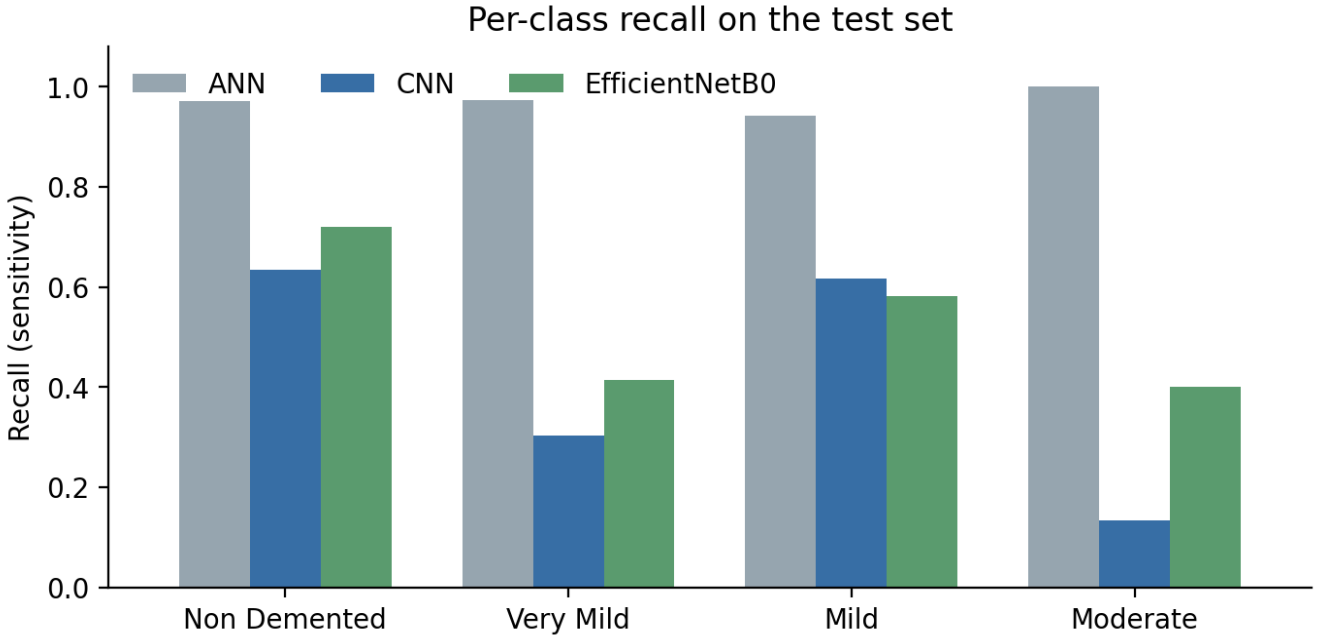


Figure 3. Recall per class on the test set.

4.2 Training behaviour

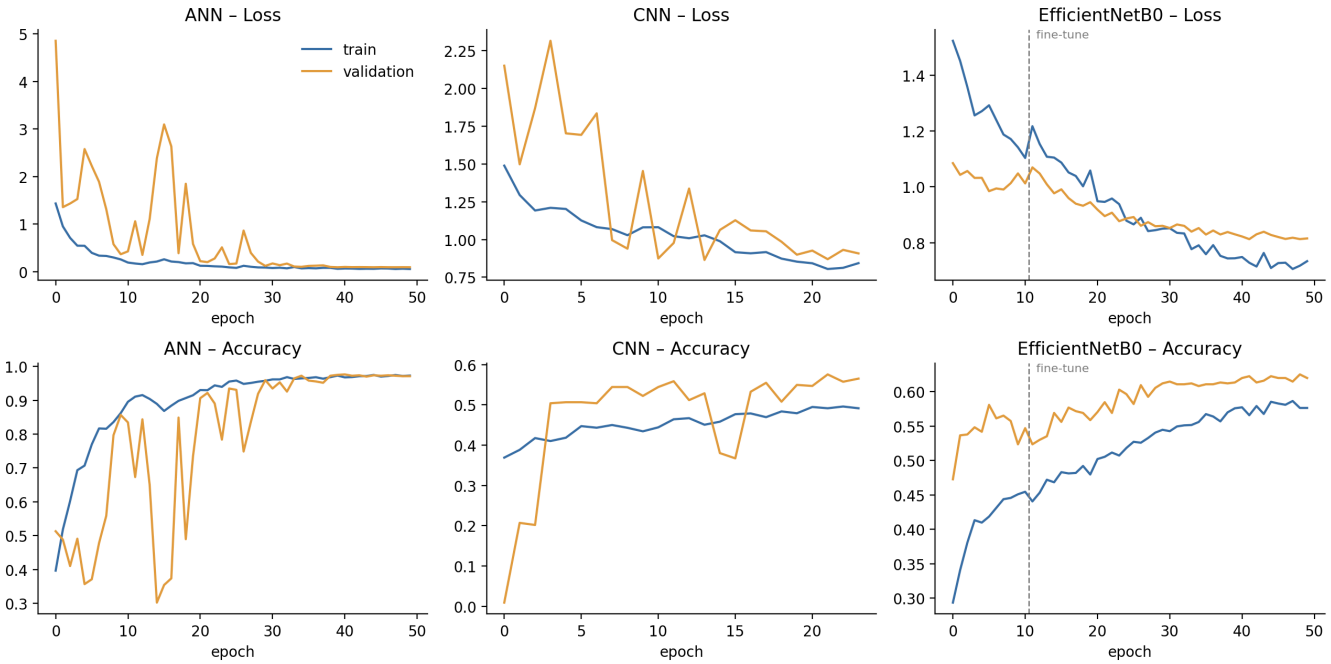


Figure 4. Loss and accuracy per epoch. The dashed line marks the start of EfficientNetB0 fine-tuning.

- **ANN.** Validation loss is very unstable for about 25 epochs, then drops close to zero. Training stopped at the 50-epoch limit with near-perfect training and validation accuracy.
- **CNN.** Training accuracy stays *below* validation accuracy (about 0.49 vs. 0.56). This is typical when strong augmentation makes the training inputs harder than the clean validation inputs. Early stopping ended training after 24 epochs.
- **EfficientNetB0.** The frozen backbone reached about 0.55 validation accuracy in 11 epochs. Fine-tuning raised it to about 0.62 and was still improving slowly when the 50-epoch budget ran out.

4.3 Statistical significance (McNemar's test)

Model A	Model B	A right, B wrong	A wrong, B right	p-value
CNN	ANN	13	604	< 1e-15
EfficientNetB0	ANN	13	500	< 1e-15
EfficientNetB0	CNN	243	139	1.4e-7

All three differences are significant at $p < 0.05$. The ANN is right on hundreds of images where the convolutional models are wrong, almost never the reverse. EfficientNetB0 is significantly better than the CNN from scratch, which supports the value of pre-training.

4.4 Grad-CAM: where does the CNN look?

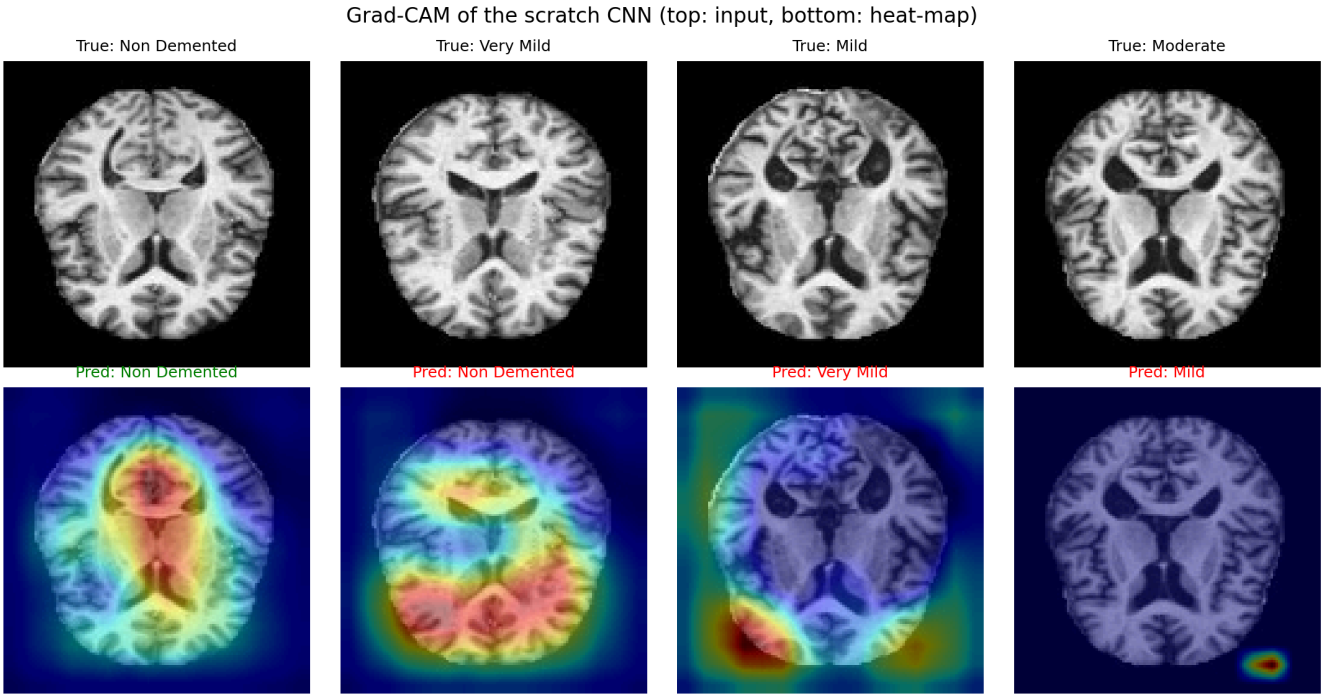


Figure 5. Grad-CAM heat-maps of the CNN (top: input slice, bottom: overlay; green = correct, red = wrong prediction).

For the correctly classified Non Demented slice, the CNN focuses on the central ventricles. This is the region where atrophy shows first, so it is a clinically sensible cue. For the three misclassified slices the attention is spread out or sits on the edge of the brain and the background. In these cases the model has not found a reliable cue.

4.5 Why does the ANN win? Leakage test and ablation

Nearest-neighbour test. A 1-NN classifier that only copies the label of the most similar training image reaches **99.8% accuracy and 0.997 macro-F1**, better than every trained network. Every test image correlates above r = 0.90 with some training image, and **98.9% correlate above r = 0.95** (median r = 0.988). For all of these, the most similar training image has exactly the same label.

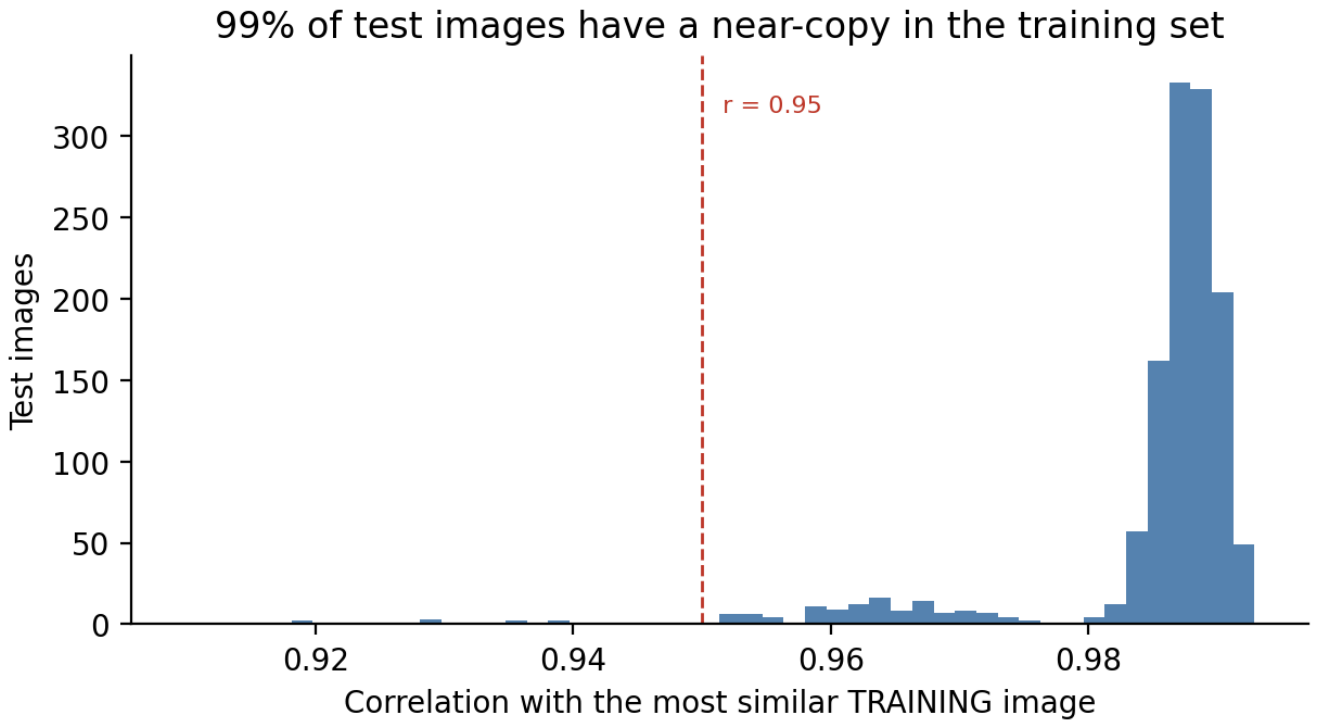


Figure 6. Correlation of each test image with its most similar training image.

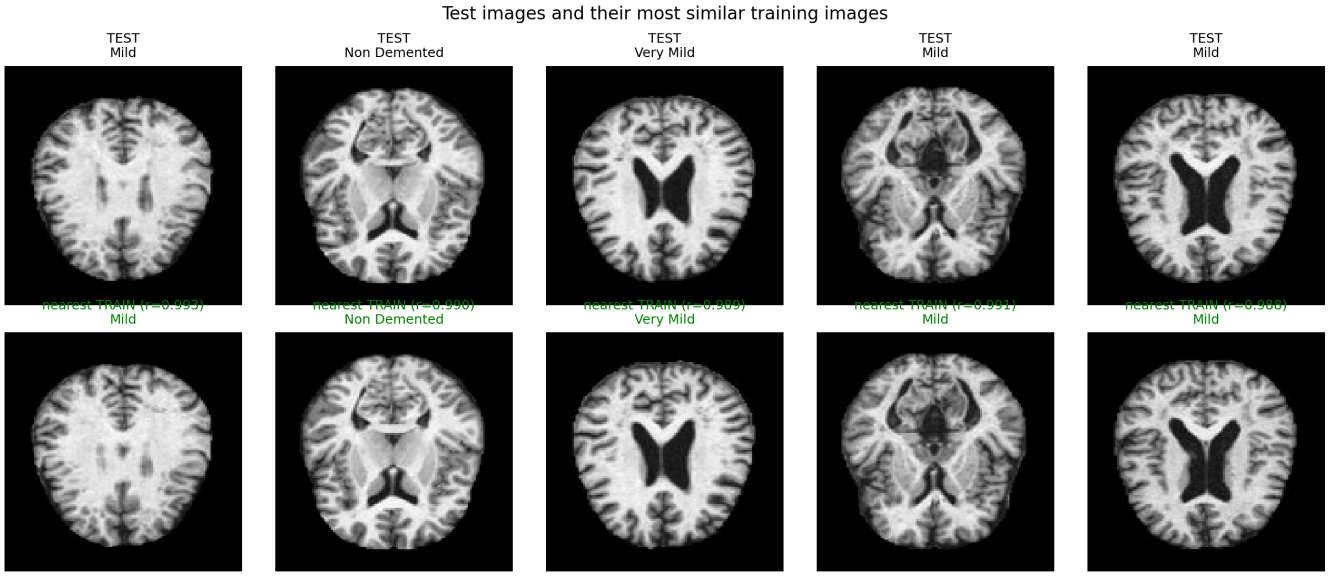


Figure 7. Test slices (top) and their most similar training slices (bottom). The pairs are visually almost identical.

The pairs in Figure 7 are adjacent slices or re-saved copies of the **same brain**. The test set therefore does not measure performance on new patients. It mostly measures how well a model recognises images it has nearly seen before.

Augmentation ablation.

Model	Augmentation	Accuracy	Macro-F1	Cohen's κ	ROC-AUC	Epochs
1-NN on raw pixels	– (no training)	0.998	0.997	–	–	–

Model	Augmentation	Accuracy	Macro-F1	Cohen's κ	ROC-AUC	Epochs
ANN	No	0.969	0.961	0.949	0.997	50
ANN	Yes	0.510	0.342	0.184	0.706	13
CNN	No	0.891	0.904	0.821	0.978	60
CNN	Yes	0.507	0.393	0.241	0.782	24
EfficientNetB0	No	0.739	0.789	0.575	0.920	53
EfficientNetB0	Yes	0.588	0.541	0.335	0.818	50
Majority class	–	0.495	0.166	0.000	–	–

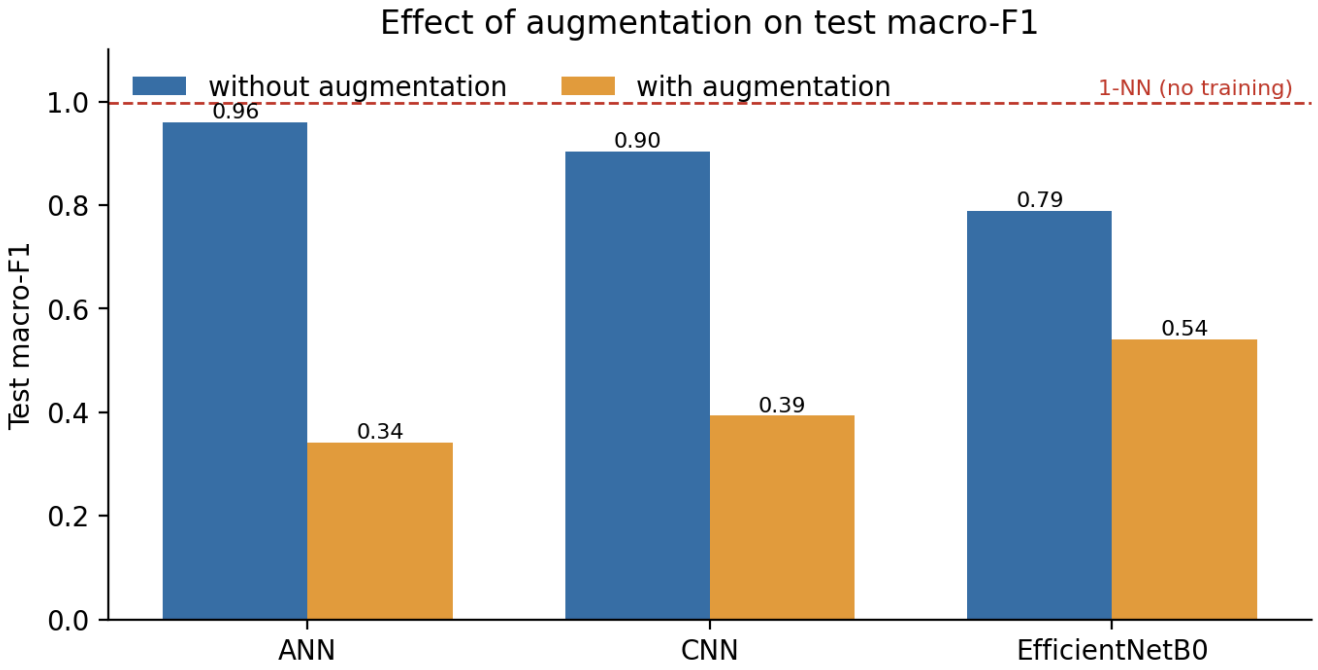


Figure 8. Test macro-F1 with and without augmentation. The dashed line is the 1-NN classifier.

The ablation confirms the explanation:

- **Without augmentation, every model scores high** (macro-F1 0.79–0.96). The ranking follows how easily a model can memorise exact pixels: the ANN with 8.5 M weights on raw pixels does best.
- **With augmentation, every model drops sharply** (macro-F1 0.34–0.54). Random shifts and rotations mean a test image no longer matches a stored pattern exactly, so the shortcut is gone.
- **Under augmentation the ranking matches our original hypothesis**: EfficientNetB0 (0.541) > CNN (0.393) > ANN (0.342) > majority (0.166). Here convolutions and pre-trained features do help.

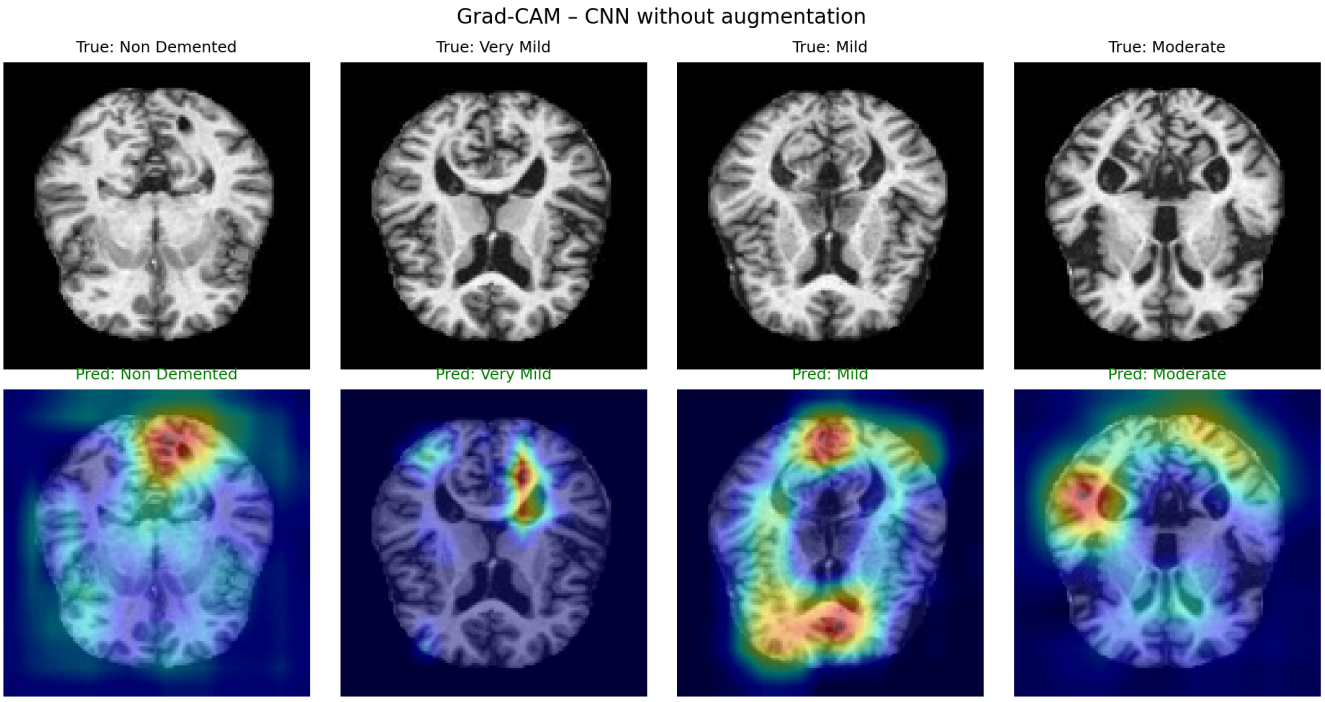


Figure 9. Grad-CAM of the CNN trained without augmentation. All four predictions are correct.

The CNN trained without augmentation gets all four Grad-CAM examples right. Yet its attention often sits on single spots near the edge of the brain rather than on the ventricles. This is consistent with matching subject-specific shapes rather than learning disease patterns.

4.6 Hypothesis vs. results

	Expected	Observed
Ranking (as first trained)	EfficientNetB0 > CNN > ANN > majority	ANN (0.961) > EfficientNetB0 (0.541) > CNN (0.393) > majority (0.166)
Ranking (all with augmentation)	EfficientNetB0 > CNN > ANN > majority	**Confirmed**: 0.541 > 0.393 > 0.342 > 0.166
Beat the majority baseline	All models	**Confirmed** for every model
Pre-training helps	EfficientNetB0 > CNN from scratch	**Confirmed** with augmentation (McNemar p = 1.4e-7)
Test score reflects new patients	Yes	**Rejected**: 99% of test images have a near-copy in training; 1-NN reaches 99.8%

The headline accuracy of about 97% is therefore **not** a realistic estimate for new patients. The augmented scores (macro-F1 0.34–0.54) are a more cautious estimate, though still not a true patient-level result.

5. Conclusion

We trained an ANN, a CNN from scratch and a pre-trained EfficientNetB0 to classify four Alzheimer's stages from 2D MRI slices. On the official split the simple ANN reached **96.9% accuracy (macro-F1 0.961)**. Our follow-up experiments show this number mainly reflects **data leakage**, not diagnostic skill. Almost every test slice has a near-identical twin in the training set, and a 1-NN lookup with no learning scores even higher (99.8%).

When augmentation removes the memorisation shortcut, all models fall to macro-F1 0.34–0.54. They then rank as expected: transfer learning > CNN > ANN > majority baseline. The most important lesson is that **a high score on a medical image benchmark must be checked for leakage before it is trusted**.

Limitations

- **No patient ids.** We could not build a patient-level split, so even the augmented scores are not clean estimates for new patients.
- **Small minority class.** With 15 Moderate test images, per-class results for that class are noisy.
- **One run per model.** Each model was trained once with one seed; we did not measure run-to-run variance.
- **Limited input.** Single 2D slices at 128 × 128 pixels, without the full 3D scan or clinical data.

Future work

1. **Patient-level split.** Use the original ADNI or OASIS data with subject ids and split by patient (Yagis et al., 2021).
2. **3D models.** Classify the whole 3D volume, for example with a 3D CNN, instead of single slices.
3. **Duplicate-aware split for this dataset.** Group near-duplicates ($r > 0.95$) into clusters and keep each cluster on one side of the split.
4. **Repeated runs.** Repeat training with several seeds and report mean ± standard deviation.
5. **Better minority handling.** Try focal loss or oversampling of the Moderate class.

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Code. Two Jupyter notebooks (Google Colab, TensorFlow/Keras): *Alzheimer_MRI_Classification.ipynb* (main experiments) and *Part2_Leakage_and_Ablation.ipynb* (leakage test and ablation).